

1.

For 95% confidence, we have $ME = 2$

$$82 - 2 < p < 82 + 2 \Rightarrow 80 < p < 84$$

A) False. The CI is constructed to estimate the population proportion, which is wrong.

The CI is constructed to estimate the population proportion and not the sample proportion.

The correct phrase would be: we are 95% confident that 82% to 84% of all Americans think it's the government's responsibility to promote equality between men and women.

B) True.

This is the correct definition of CI for this question.

We are 95% confident that 82% to 84% of all Americans think it's the government's responsibility to promote equality between men and women.

C) True.

For 95% confidence intervals means that if we take many random samples and calculate confidence interval for each sample, 95% of these intervals contain the true population proportion.

D) True.

$$\frac{ME_1}{ME_2} = \frac{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1}}}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_2}}} = \frac{2}{1} \Rightarrow \sqrt{\frac{n_2}{n_1}} = 2 \Rightarrow n_2 = 4n_1$$

E) True.

we are 95% confident that Americans think it's the government's responsibility to promote equality between men and women.

2. It was stated that the question is incomplete and there is no need to answer it.

3.

a)

Name	Yes	No	Total	$\hat{p}$	p_{pool}
Lopinavir	26	110	120	$\frac{26}{120}$	$\frac{26}{240}$
Nevaripine	10	94	120	$\frac{10}{120}$	

b) $H_0: P_{\text{nevaripine}} = P_{\text{Lopinavir}}$: There is no significant difference.

$H_A: P_{\text{nevaripine}} \neq P_{\text{Lopinavir}}$: There is a significant difference.

c)

we check condition:

- Independence:

- within groups: sampled observations are independent within each group.

- The women have been randomly assigned to each treatment group. ✓

- sampling is without replacement, and the number of the sample is less than 10% population of women. ✓

- between groups: the two groups of treatment are independent of each other. ✓

- Sample size / Skew: Each sample should meet the success - Failure condition:

- $n_1 P_{\text{pool}} \geq 10$ and $n_1 (1 - P_{\text{pool}}) \geq 10 \Rightarrow$

- $120 \times \frac{36}{240} = 18 > 10$ and $120 \times \frac{204}{240} = 102 > 10$ ✓

- $n_2 P_{\text{pool}} \geq 10$ and $n_2 (1 - P_{\text{pool}}) \geq 10 \Rightarrow$

- $120 \times \frac{36}{240} = 18 > 10$ and $120 \times \frac{204}{240} = 102 > 10$ ✓

we can use normal model.

$$P_{\text{pool}} = \frac{36}{240}$$

$$SE = \sqrt{\frac{P_{Pool}(1-P_{Pool})}{n_1} + \frac{P_{Pool}(1-P_{Pool})}{n_2}} = \sqrt{\frac{\frac{36}{240} \times (1 - \frac{36}{240})}{120} + \frac{\frac{36}{240} \times (1 - \frac{36}{240})}{120}}$$

$$= 0.04609772$$

$$Z = \frac{\text{observation} - \text{null}}{SE} = \frac{\frac{26}{120} - \frac{10}{120} - 0}{0.04609772} = 2.892406$$

$$P(|Z| > 2.892406) = 0.001911518$$

`pnorm(2.892406, lower.tail = FALSE)`
 0.001911518

$P\text{-value} < \alpha = 0.05 \Rightarrow$ Therefore, There is strong evidence to reject H_0 . This means, there is a difference in the proportion of virologic failures among those who take Nevirapine and those who take Lopinavir.

4.

	Sample size (n _i)	Approve law (%)	Disapprove law (%)	Other
"People who cannot afford it will receive financial help from the government" is given second	771	47	49	4
"People who do not buy it will pay a penalty" is given second	732	34	63	3

we check conditions:

- Independence:

- within groups: the observations are independent, both within the samples and between the samples.

- the statements in brackets were randomly assigned to each respondent. ✓

- Sampling is without replacement, and the number of the sample is less than 10% population. ✓

- between groups: the two groups of the respondent are independent of each other. ✓

- Sample size / Skew: Each sample should meet the success-failure condition:

- $n_1 \hat{p}_1 \geq 10$ and $n_1 (1 - \hat{p}_1) \geq 10 \Rightarrow$

$$771 \times 0.47 = 362.37 > 10 \text{ and } 771 \times 0.53 = 408.63 > 10 \quad \checkmark$$

- $n_2 \hat{p}_2 \geq 10$ and $n_2 (1 - \hat{p}_2) \geq 10 \Rightarrow$

$$732 \times 0.34 = 248.88 > 10 \text{ and } 732 \times 0.66 = 483.12 > 10 \quad \checkmark$$

with these conditions verified, the normal model can be used for the point estimate of the difference in support.

$$SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} = \sqrt{\frac{0.47(1-0.47)}{771} + \frac{0.34(1-0.34)}{732}}$$

$$= 0.02509$$

For 95% confidence, we have $z^* = 1.6448$

R

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qnorm(0.95)
1.644854
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$$\hat{p}_2 - \hat{p}_1 \pm z^* \times SE = (0.49 - 0.34) \pm 1.6448 \times 0.02509$$

$$\Rightarrow (0.08873, 0.1712)$$

we are 90% confident that the approval rating for the 2010 healthcare law changes is between 9% and 17% due to the ordering of the two statements in the survey question.

5.

Options	Rock	Scissors	Paper	Total
Time played	43	21	35	99
Expected	33	33	33	99

our hypothesis would look like this:

$$H_0: P_{\text{Rock}} = P_{\text{scissors}} = P_{\text{paper}} :$$

All options are favored equally. ($p = 1/3$)

H_A : Some option are favored more than others options.

we check conditions:

- Independence: Sampled observations are independent. ☒
 $n < 10\%$ of population ☒

And each case only contributes to one cell. ☒

- Sample Size: Expected value is equal to np , where the proportion for each is 0.33 assuming an equal chance to pick each one, and the sample size is 99 ($43 + 21 + 35$). All the expected values will be 33, which is greater than 5. Therefore, we have a large enough sample. ($n \geq 5$) ☒

chi-square GOF :

$$\chi^2 = \frac{(43 - 33)^2}{33} + \frac{(21 - 33)^2}{33} + \frac{(35 - 33)^2}{33} = 7.515152$$

$$df = K - 1 = 3 - 1 = 2$$

$$p\text{-value} = 0.023$$

R

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pchisq(7.515152, 2, lower.tail = FALSE)
0.02334025
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p -value is less than $\alpha = 0.05$. Therefore, There is strong evidence to reject H_0 . This means, some of the options are favored more than others options.

6.

	Ketchup	Mustard	Relish	Total
Male	15 (19.2)	23 (20.16)	10 (8.64)	48
Female	25 (20.8)	19 (21.84)	8 (9.36)	52
Total	40	42	18	100

our Hypothesis is:

H_0 : Gender and condiment are independent.

H_A : Gender and condiment are dependent.

we check conditions:

- Independence: Sampled observations are independent ✓ and without replacement and less than 10% of the ✓ population. And each case only contributes to one cell. ✓
- Sample Size: None of the expected counts in the table are less than 5. ✓

Conditions verified ✓

$$\text{Expected count} = \frac{(\text{row total})(\text{column total})}{\text{total table}}$$

$$E_{\text{male, Ketchup}} = \frac{40 \times 48}{100} = 19.2$$

$$E_{\text{male, Mustard}} = \frac{42 \times 48}{100} = 20.16$$

$$E_{\text{male, Relish}} = \frac{18 \times 48}{100} = 8.64$$

$$E_{\text{female, Ketchup}} = \frac{40 \times 52}{100} = 20.8$$

$$E_{\text{female, Mustard}} = \frac{42 \times 52}{100} = 21.84$$

$$E_{\text{female, Relish}} = \frac{18 \times 52}{100} = 9.36$$

$$\chi^2 = \frac{(15 - 19.2)^2}{19.2} + \frac{(23 - 20.16)^2}{20.16} + \frac{(10 - 8.64)^2}{8.64} + \frac{(25 - 20.8)^2}{20.8} + \frac{(19 - 21.84)^2}{21.84} + \frac{(8 - 9.36)^2}{9.36} =$$

$$= 2.947891$$

$$df = 3 - 1 = 2$$

$$p\text{-value} = 0.229$$

R:

pchisq(2.947891, 2, lower.tail = FALSE)
0.2290201

with a p -value greater than 5%, Since p -value is high we fail to reject H_0 . This means, we conclude that there is not enough evidence in the data to suggest that gender and preferred condiment are dependent.

7. H_0 : Absence occurs with equal frequencies during a five day workweek.

H_A : Absence does not occur with equal frequencies during a five-day workweek.

we check conditions:

- Independence: Sampled observations are independent. And each case only contributes to one cell.
- Sample size: Expected value is 14 which is larger than 5.

	monday	Tuesday	wednesday	Thursday	Friday
Number of Absences	18	16	10	10	16
Expected	14	14	14	14	14

Chi-squared test for given probabilities

data: absence_counts
X-squared = 4, df = 4, p-value = 0.406

$$p\text{-value} = 0.406 > \alpha$$

with a p -value greater than 5%, since p -value is high we fail to reject H_0 . Therefore, we conclude that there is not enough evidence in the data to suggest that absence does not occur with equal frequencies during a five-day workweek.

8.

H_0 : The students' smoking habit is independent of their exercise level.

H_A : The students' smoking habit is dependent on their exercise level.

we check conditions:

- Independence: Sampled observations are independent. And each case only contributes to one cell. ☒
- Sample size: each particular scenario does not have at least 5 cases. Therefore, this condition is False.

The second condition is not verified for doing the test.

I assume the conditions are met:

Pearson's Chi-squared test

data: contingency_table

X-squared = 5.4885, df = 6, p-value = 0.4828

p-value = 0.4828. with a p-value greater than 5%, Since p-value is high we fail to reject H_0 . Therefore, we conclude that there is not enough evidence in the data to suggest that the smoking habit is dependent on the exercise level of the students.